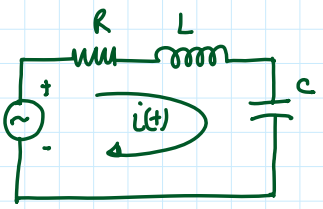
Lecture 21: Design of Passive Filters



In general, $\bar{I} = \frac{\bar{V}}{R + j(\omega L - \frac{1}{\omega C})} = \frac{V/R}{1 + j\left(\frac{\omega L}{R} - \frac{1}{\omega C R}\right)}$

$$= \frac{V/R}{1 + j\left(\frac{\omega_0 L}{R}\frac{\omega}{\omega_0} - \frac{1}{\omega_0 C}\frac{\omega_0}{R\omega}\right)} \qquad \omega_0 L = \frac{1}{\omega_0 C}$$

$$= \frac{V/R}{1 + j\frac{\omega_0 L}{R}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)}$$

$$\bar{I} = \frac{V/R}{\sqrt{1 + \left(\frac{\omega_0 L}{R}\right)^2\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2}} \cdot \angle - \tan^{-1}\frac{\omega_0 L}{R}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)$$

$= I \angle \phi_I$

Bandwidth, ω_b is defined range of frequencies for which the power delivered to the load is half of the power at resonance.

$P_0 = I_0^2 * R$

$P = \frac{P_0}{2} = I_1^2 R = \frac{I_0^2 R}{2}$

$I_1 = \frac{I_0}{\sqrt{2}} = 0.707\, I_0$

Lower cut-off frequency:

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

Higher cut-off frequency

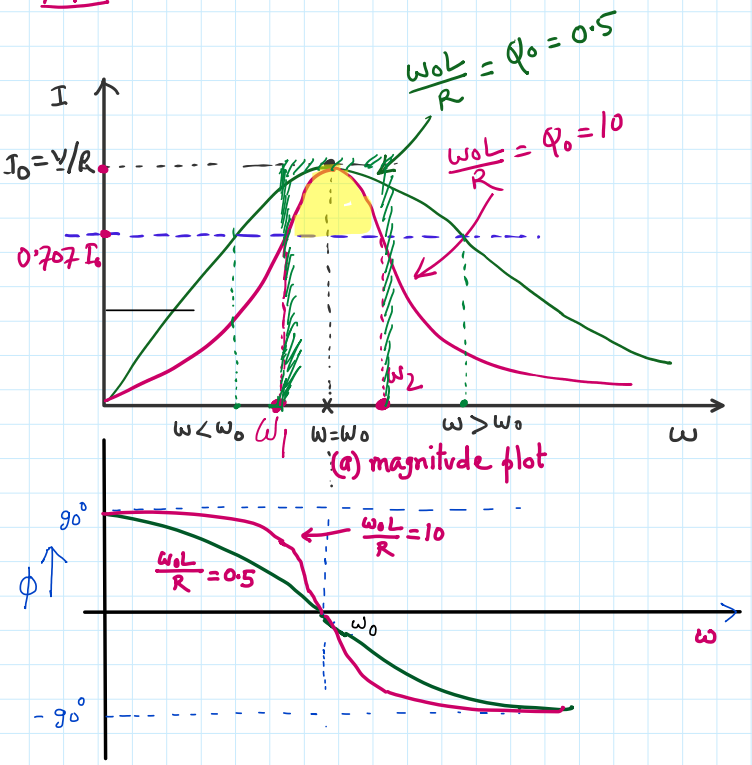
$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

Bandwidth:

$$\boxed{\omega_b = \omega_2 - \omega_1 = \frac{R}{L}}$$

Quality Factor:

$$Q_0 = \frac{\omega_0}{\omega_b} = \frac{\omega_0 L}{R} = \frac{X_{L0}}{R}$$



(b) Phase plot

ω_1 = lower cut-off frequency
ω_2 = higher cut-off frequency

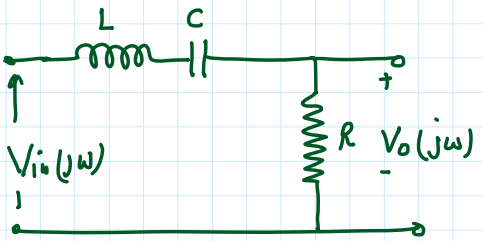
Filter Characteristics → ① Band Pass Filter
② Band Rejection Filter
③ Low Pass Filter
④ High Pass Filter

From this discussion we realize that there is output which is frequency

From this discussion we realize that there is output, which is frequency dependent, hence the term frequency response.

The magnitude plot of current response shows that, the current is dominantly present if $\omega_1 < \omega < \omega_2$. Now if we take output across resistive element, as shown in the figure.

The output voltage, V_o is given by,



$$\bar{I}R = \frac{V/R \times R}{\sqrt{1+\left(\frac{\omega_0 L}{R}\right)^2\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)^2}} \cdot \angle -\tan^{-1}\frac{\omega_0 L}{R}\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)$$

$$\bar{V}_o = \bar{I}R = \frac{\bar{V}_{in}}{\sqrt{1+\left(\frac{\omega_0 L}{R}\right)^2\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)^2}} \angle -\tan^{-1}\frac{\omega_0 L}{R}\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)$$

$$\left|\frac{\bar{V}_o}{\bar{V}_{in}}(j\omega)\right| = \frac{1}{\sqrt{1+\left(\frac{\omega_0 L}{R}\right)^2\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)^2}}$$

$\Rightarrow$ Band Pass Filter Characteristics

The above is know as Transfer function.

(A) When $\omega = \omega_0$

$$\left|\frac{\bar{V}_o}{\bar{V}_{in}}\right| = 1$$

(B) When $\omega = \omega_1$; $\left|\frac{\bar{V}_o}{\bar{V}_{in}}\right| = 0.707$

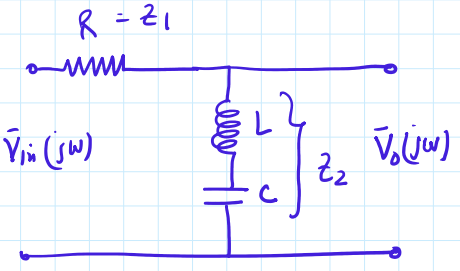
(C) When $\omega = \omega_2$, $\left|\frac{\bar{V}_o}{\bar{V}_{in}}(j\omega)\right| = 0.707$

(D) For $\omega < \omega_1$ & $\omega > \omega_2$

$$\left|\frac{V_o}{\bar{V}_{in}}(j\omega)\right| < 0.707$$

This satisfies the characteristic of Band Pass Filter as it allows signal to pass within $\omega_1 < \omega < \omega_2$ frequency.

Band Rejection Filter:



$$\frac{\bar{V}_o}{\bar{V}_{in}}(j\omega) = T(j\omega) = \frac{\left(j\omega L - \frac{j}{\omega C}\right)}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

$$= \frac{j(\omega^2 LC - 1)/\omega C}{\omega RC + j(\omega^2 LC - 1)/\omega C}$$

$$\frac{\bar{V}_o}{V_{in}} = \frac{Z_2}{Z_1 + Z_2}$$

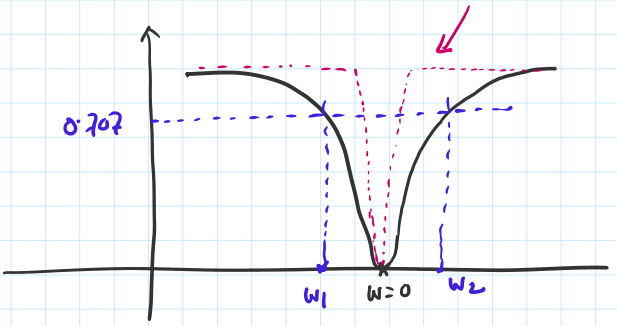
$$\frac{WRC + j(W^2LC - 1)}{WC} \Big/ V_{in}$$

$$= \frac{j(W^2LC-1)}{WRC + j(W^2LC-1)} = \frac{1}{1 - j\dfrac{WRC}{(W^2LC-1)}}$$

$$T(jW) = \frac{1}{\left[1 + \dfrac{(WRC)^2}{(W^2LC-1)^2}\right]^{1/2}} \Big/ \tan^{-1}\frac{WRC}{(W^2LC-1)}$$

$$= |T(jW)| \Big/ T(jW)$$

$$|T(jW)|_{W=W_0} = 0$$



$W < W_1$ } Signal is passed
$W > W_2$ }

$W_1 < W < W_2$ } signal is rejected } Band Rejection Filter